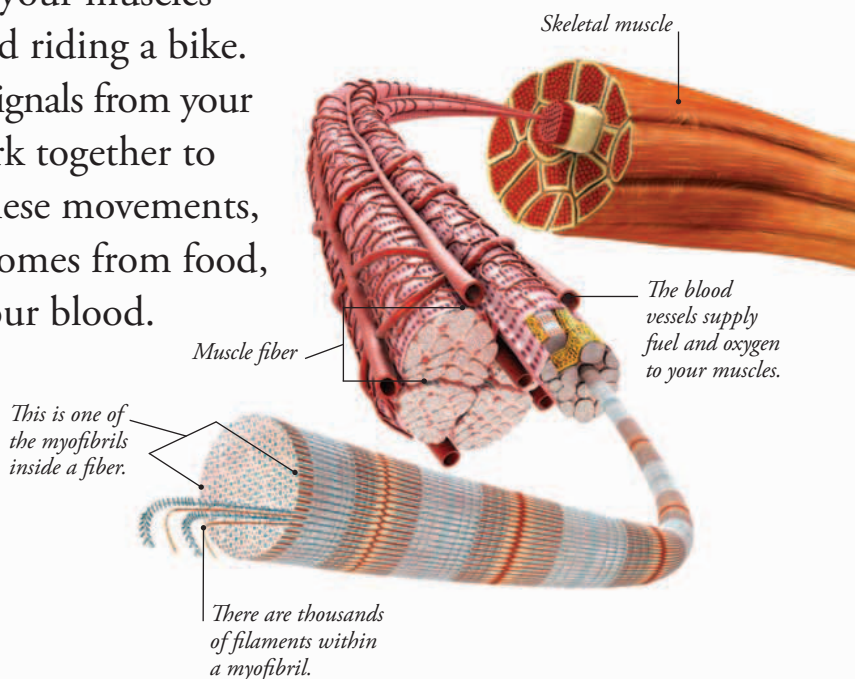


Body movers

Every movement you make depends on your muscles—from blinking and eating, to walking and riding a bike. Your movements are controlled by nerve signals from your brain, which instruct the muscles to work together to produce precise movements. To make these movements, your muscles need energy. This energy comes from food, which is delivered to your muscles by your blood.

HOW MUSCLES WORK

Skeletal muscles, such as those found in your arms, are formed by bundles of muscle fibers. Each muscle fiber is made of smaller rod-shaped fibers called myofibrils, and within these are superthin fibers known as filaments. When the filaments slide together, all the muscle fibers shorten, so the whole muscle contracts, pulls on your bones, and moves your arm.



TEAMWORK

When a muscle is stimulated by a nerve signal, it contracts and gets shorter. It cannot make itself longer. This means that muscles can pull, but not push. Many muscles, such as the those in your upper arm, are arranged in pairs that work against each other, pulling in different directions. For instance, the biceps pulls your forearm up, and the triceps pulls it down again.

